

The Iwasawa Decomposition of $GL_n(\mathbb{R})$

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Motivation

The Iwasawa decomposition expresses every invertible real $n \times n$ matrix as a product of three matrices, each drawn from a distinguished subgroup. Lang treats it in Appendix II of *Linear Algebra* (3rd ed., 1987), titled “Iwasawa Decomposition and Others.” The appendix’s stated purpose is to introduce the language of group theory through the running example of SL_n (the group of $n \times n$ real matrices with determinant 1), which had not been used in the rest of the book; Lang chooses the Iwasawa decomposition as the vehicle, describing it explicitly as a way to “express Theorem 2.1 of Chapter V in the context of groups and subgroups.” That theorem is the Gram–Schmidt orthonormalization theorem, so the Iwasawa decomposition is, for Lang, the group-theoretic restatement of Gram–Schmidt: the assertion that the product map $U \times A \times K \rightarrow SL_n$ is a bijection, where U is the upper-unipotent subgroup, A the positive diagonal subgroup, and K the orthogonal subgroup. From there the appendix moves on to the broader content suggested by “Others”: the actions of $SL_2(\mathbb{R})$ and $SL_2(\mathbb{C})$ on the hyperbolic upper half-plane and upper half-space, the conjugation action with the first rudiments of Lie algebras, and the complementary polar decomposition $G = KAK$. The Iwasawa decomposition itself is the matrix face of a more general phenomenon in Lie theory: any real semisimple Lie group G admits a diffeomorphism $G \cong K \times A \times N$, where K is a maximal compact subgroup, A a maximal split abelian subgroup, and N a maximal unipotent subgroup. The formalization here works with the slightly larger group $GL_n(\mathbb{R})$ (no determinant constraint); the proof carries over from Lang’s argument with no essential change.

Statement and proof

Write M_n for the type of $n \times n$ real matrices. A matrix is *positive diagonal* if it is diagonal with strictly positive entries, and *upper unipotent* if it is upper triangular with ones on the diagonal.

Theorem (Iwasawa decomposition for $GL_n(\mathbb{R})$). *For every $g \in M_n$ with $\det g \neq 0$ there exist unique $k, a, u \in M_n$ such that $g = k \cdot a \cdot u$, where k is orthogonal, a is positive diagonal, and u is upper unipotent. (Lang states the bijection in the form $g = uak$; the two orderings are equivalent by inversion since each of U, A, K is closed under inverses.)*

Existence via Gram–Schmidt

Following Lang’s Appendix II, view column i of g as $g^{(i)} \in \mathbb{R}^n$ with $(g^{(i)})_j = g_{ji}$. Since $\det g \neq 0$, the columns are linearly independent. Apply Gram–Schmidt: produce vectors $\tilde{e}_i$, the components of $g^{(i)}$ orthogonal to the span of the earlier columns, and let $e_i = \tilde{e}_i / \|\tilde{e}_i\|$. Form Q with i -th column

e_i ; orthonormality of (e_i) is exactly $Q^T Q = I$, so Q is orthogonal. Set $R = Q^T g$, equivalently $R_{ij} = \langle e_i, g^{(j)} \rangle$. Two facts about R follow from Gram–Schmidt: (i) for $j < i$, the i -th Gram–Schmidt vector is orthogonal to $g^{(j)}$ (which lies in the span of earlier columns), so $R_{ij} = 0$ and R is upper triangular; (ii) expanding $g^{(i)}$ as $\tilde{e}_i$ plus projections onto the earlier $\tilde{e}_k$ and pairing with $\tilde{e}_i$ collapses to $\langle \tilde{e}_i, g^{(i)} \rangle = \|\tilde{e}_i\|^2$, hence $R_{ii} = \|\tilde{e}_i\| > 0$. Now split $R = D \cdot U$ with D the diagonal of R and $U = D^{-1}R$. Then D is positive diagonal, and U is upper unipotent ($U_{ii} = R_{ii}^{-1} \cdot R_{ii} = 1$). Combining, $g = QR = Q(DU)$, so $k = Q$, $a = D$, $u = U$ is the required factorization.

Uniqueness via a key lemma

Uniqueness rests on:

Key lemma. *If M is orthogonal, upper triangular, and has positive diagonal, then $M = I$.*

Proof. Orthogonality gives $M^{-1} = M^T$. The inverse of an upper-triangular invertible matrix is upper triangular, so M^T is upper triangular, equivalently M is lower triangular. With both upper and lower triangular, M is diagonal. From $MM^T = I$ each $M_{ii}^2 = 1$, and positivity selects $M_{ii} = 1$. $\square$

To finish uniqueness, suppose $g = k_1 a_1 u_1 = k_2 a_2 u_2$ and set $M = k_2^T k_1$. Then M is orthogonal (product of orthogonals), and rearranging gives $M = a_2 u_2 u_1^{-1} a_1^{-1}$, which is upper triangular with positive diagonal (the diagonal of a product of upper-triangular matrices is the product of diagonals; positivity follows from positivity of a_1, a_2 and unipotence of u_1, u_2). The lemma forces $M = I$, so $k_1 = k_2$. Cancelling gives $a_1 u_1 = a_2 u_2$, hence $a_1 = a_2(u_2 u_1^{-1})$; comparing diagonals forces $a_1 = a_2$ and the unipotent factor to be the identity, so $u_1 = u_2$.

The formalization landscape in Mathlib

Mathlib provides each structural ingredient required by the proof but does not package the decomposition itself. `Mathlib.LinearAlgebra.UnitaryGroup` defines the orthogonal group on real matrices; `Mathlib.Analysis.InnerProductSpace.GramSchmidtOrtho` provides `gramSchmidt` and `gramSchmidtNormed` together with orthogonality, span, and nonvanishing lemmas; and `Mathlib.LinearAlgebra.Matrix.Block` provides the `BlockTriangular` predicate together with the result that the inverse of an upper-triangular invertible matrix is upper triangular. The closest existing matrix factorization in the library is the LDL decomposition in `Mathlib.Analysis.Matrix.LDL`, useful as a structural template, but neither the Iwasawa decomposition nor the closely related QR factorization (numerical linear algebra’s name for the same data; see Trefethen–Bau) is packaged anywhere in Mathlib. The accompanying repository https://github.com/CuteSurtr/Iwasawa_Decomposition contains a complete Lean 4 / Mathlib formalization corresponding to the argument above; the full formal proof, with explicit references to every Mathlib lemma used, appears in the repository’s `README.md`.

References

1. S. Lang. *Linear Algebra*, 3rd edition. Springer, 1987. (Appendix II, “Iwasawa Decomposition and Others.”)
2. L. N. Trefethen, D. Bau. *Numerical Linear Algebra*. SIAM, 1997. (Lectures 7–8 on QR.)
3. J. Miao, K. Buzzard, A. Bentkamp. `Mathlib.Analysis.InnerProductSpace.GramSchmidtOrtho`.
4. A. Bentkamp. `Mathlib.Analysis.Matrix.LDL`.
5. The Mathlib community. <https://leanprover-community.github.io/>.